

Level-2, Task-2

Wine Quality Classification using Machine Learning

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Track: Data Analytics

1. Introduction

Wine quality is influenced by several physicochemical properties such as acidity, alcohol content, sulphates, density, chlorides, and sulfur dioxide levels. Predicting wine quality using machine learning helps automate the evaluation process, reducing the need for manual assessment while providing consistent and data-driven results.

The objective of this project was to build and compare multiple classification models capable of predicting whether a wine belongs to the **Good** or **Bad** quality category based on its chemical characteristics. Three machine learning algorithms—Random Forest, Support Vector Classifier (SVC), and Stochastic Gradient Descent (SGD)—were trained and evaluated to determine the most suitable model for deployment.

2. Dataset Description

The Wine Quality dataset consists of physicochemical measurements collected from wine samples. Each record contains several numerical features describing the chemical composition of the wine.

Features Used

- Fixed Acidity
- Volatile Acidity
- Citric Acid
- Residual Sugar
- Chlorides
- Free Sulfur Dioxide
- Total Sulfur Dioxide
- Density
- pH
- Sulphates
- Alcohol

The original target variable, **Quality**, ranged from **3 to 8**. To simplify the classification task and reduce the impact of class imbalance, the quality scores were converted into two categories:

- **Bad Quality (0):** Quality < 6
 - **Good Quality (1):** Quality ≥ 6
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3. Exploratory Data Analysis

The dataset was initially inspected to understand its structure, statistical properties, and quality distribution. Missing values were checked and the unnecessary **Id** column was removed before model training.

Several exploratory data analysis techniques were performed, including:

- Distribution plots for all numerical features
- Correlation heatmap
- Quality class distribution analysis

The correlation analysis showed that **alcohol** had the strongest positive relationship with wine quality, while **volatile acidity** exhibited a negative relationship. Features such as sulphates and total sulfur dioxide also contributed significantly to predicting wine quality.

The class distribution indicated that the original quality scores were imbalanced, with middle-quality wines appearing much more frequently than the extreme quality scores. Converting the target into binary classes helped simplify the learning problem and improved model performance.

4. Model Development

The dataset was divided into training and testing sets using an **80:20 stratified split**, ensuring that both datasets preserved the original class proportions.

Feature scaling using **StandardScaler** was applied to the SGD and SVC models because these algorithms are sensitive to feature magnitudes. Random Forest was trained using the original feature values, as tree-based algorithms do not require feature scaling.

The following classification algorithms were implemented:

- Random Forest Classifier
- Support Vector Classifier (SVC)
- Stochastic Gradient Descent (SGD) Classifier

Each model was evaluated using Accuracy, Precision, Recall, F1-score, Classification Report, and Confusion Matrix.

5. Results

The comparison of model performance is presented below.

Model	Accuracy
Random Forest	80.35%
Support Vector Classifier (SVC)	78.60%
Stochastic Gradient Descent (SGD)	62.88%

Among the three models, **Random Forest** achieved the highest accuracy and the best balance between precision and recall. SVC also produced competitive results but performed slightly below Random Forest. Although SGD achieved relatively high recall, its lower precision and overall accuracy indicated that it generated more incorrect positive predictions.

Feature importance analysis from the Random Forest model showed that **alcohol** was the most influential predictor of wine quality, followed by **sulphates**, **total sulfur dioxide**,

volatile acidity, and **density**. These findings suggest that wine quality depends on a combination of chemical properties rather than a single characteristic.

6. Conclusion

This project successfully demonstrated the application of machine learning techniques for wine quality classification. Exploratory data analysis provided valuable insights into the dataset, while feature engineering helped reduce the effects of class imbalance by transforming the original quality scores into binary classes.

Among the three evaluated classifiers, **Random Forest** proved to be the most effective model, achieving an accuracy of **80.35%** along with the best overall balance of precision, recall, and F1-score. Additionally, its feature importance analysis improved model interpretability by identifying the most significant chemical properties influencing wine quality.

Overall, Random Forest is the most suitable model for deployment because of its strong predictive performance, robustness to non-linear relationships, and ability to provide meaningful insights into feature contributions. Future work could focus on hyperparameter tuning, cross-validation, multiclass quality prediction, and advanced techniques such as SMOTE to further improve classification performance.